

Understand The Problem

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Buck Converter: $3.3V \rightarrow 1.2V$

Supply ripple : less than 100mV peak to peak

Switching node ringing value : less than 1.5V (due to parasitic inductances)

Soc can consume a peak current of 10mA. But, during sleep mode, it must consume 10μA.
Peak efficiency should be at least 90%.

Design should be able to reject an interference signal @ 2kHz from nearby by ~40dB.
All components from 180BCD

Duty Cycle calc.

$$\frac{V_o}{V_{in}} = D \Rightarrow \frac{1.2}{3.3} = 0.3636$$

$$t_{on} = DT \text{ . Let } f = 2MHz \Rightarrow T = \frac{1}{2 \times 10^6} = 500n \Rightarrow t_{on} = DT = 181.8ns$$
$$t_{off} = 318.2ns$$

Ripple Current Calculation:

Let's assume $L = 4\mu H$. [sounds like a good trade off betⁿ ripple current & Area]

$$\text{On-time: } V_L = V_{in} - V_o = 3.3 - 1.2 = 2.1V$$

$$\text{off-time: } V_L = -V_o = -1.2V$$

$$\text{On-time } \Delta I_L = \frac{(V_{in} - V_o) t_{on}}{L} = \frac{2.1 \times 181.8n}{4\mu} = 95.445mA$$

$$\text{off-time } \Delta I_L = \frac{(-V_o) t_{off}}{L} = \left| \frac{(-1.2) \times 318.2}{4\mu} \right| = 95.445mA$$